

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. INTRODUCTION :

To overcome the limitations of rigid, pre-programmed fitness templates, we propose **FitBuddy**. This comprehensive web platform acts as an automated virtual personal trainer, integrating modern web routing with advanced Generative AI to deliver highly personalized health guidelines.

3. CORE SYSTEM CAPABILITIES:

The proposed solution is designed to meet several key project objectives:

- **Responsive Data Collection:** The development of a responsive web application that efficiently collects user fitness profiles and their specific physical goals.
- **Dynamic Content Generation:** The integration of the Google Gemini API to instantly create customized, day-by-day workout plans tailored to the user.
- **Data Persistence:** The implementation of a relational database system to persistently store both the user profile data and the AI-generated workout plans, enabling future retrieval and administrative monitoring.
- **System Reliability:** The inclusion of seamless fallback mechanisms, specifically timeout handling, which ensures a continuous user experience even during periods of API network latency.

4. TECHNOLOGY STACK PROPOSAL:

To execute this solution effectively, the system will be built using Python 3. The backend will be powered by the FastAPI framework and hosted on a Uvicorn server. For data persistence, SQLite will be utilized alongside SQLAlchemy as the Object-Relational Mapper (ORM). The artificial intelligence component will be driven by the Google Gemini API, specifically utilizing the `gemini-3.5-flash-lite` model.

Finally, the frontend user interface will be constructed using Jinja2 Templates, HTML, and CSS.